Isoliner · Structural lines: crests and toes

A cheat sheet for tools 2.13-2.15 · QGIS 3.16+ · Isoliner plugin · no GRASS or SAGA

Pipeline: from a dense survey to crest-toe forms

Step 1. Terrain. A dense survey is needed (UAV, laser scanning). A sparse survey has no crest in the data at all. Use a **ground** raster, not a DSM. Vegetation and machinery are removed by classification before Isoliner. Cell from density: 20-25 cm at 20 points per m², 15 cm at 40. Without your own data use **2.15 Example open pit (demo)**.

Step 2. Candidates - 2.13 Crest and toe candidates. Input: the terrain of step 1. The noise cut-off is 0.5 m, the minimum length 10 cells, the probe base 8 cells. Output: lines with the fields **drop** (across the line, m), **length_m**, **slope_deg** and **kind** (brow or toe). The layer arrives coloured by the drop.

Step 3. Selection on the map. The log holds the percentiles of the drop over all candidates. Set a layer filter `"drop" > 1.0` and move the number while watching the map. There is no need to rerun 2.13, and the threshold you choose is your criterion of a bench.

Step 4. Forms - 2.14 Crests and toes into work. Crests and toes already drawn (a topographic deliverable, a mine survey)? Steps 1-3 are not needed, start here. Input: the candidates of step 2 (the whole layer, the drop cut-off is available here too) and the same DEM. Keep the descent path limit close to the width of the face. Output: **Top** and **Bottom** with a shared **link** field, elevations taken off the DEM into the Z geometry, plus a layer of unpaired lines with the reason.

Step 5. Where the forms go. The forms go into surface building between lines, into 2.03 as structural constraints, or to an export into AutoCAD and Credo as 3D lines.

A check on the demo. Lay the candidates over the true lines of 2.15. They must lie along them, break on the ramp arc, and give two crests with no toes at the ditch.

Tool reference: input, output, key parameters

Tool	Input	Output	Key parameters
2.13 Crest and toe candidates	DEM (dense, ground)	lines: kind, drop, length_m, slope_deg	the 0.5 m drop cut-off removes ruts. Length 10 cells. The probe base is 8 cells: the drop is measured within it, not across the whole bench
2.14 Crests and toes into work	candidates of 2.13, the same DEM	Top and Bottom (LineStringZ, kind, link) + the unpaired with a reason	descent path limit 50 m, by the width of the face. The share of agreeing probes 0.4 is in the advanced parameters
2.15 Example open pit (demo)	nothing	DEM + true lines (kind, link)	“What to build”: a pit with berms, a simple bench or a dump, the rest with “my own”. The RNG seed repeats the run

How it works

- **The evidence of a break** is the magnitude of the slope gradient, not the slope itself. On an even face, however steep, the slope is constant and the evidence is small. The evidence is large where the slope changes, that is on a crest and on a toe.
- **The sign of the curvature** splits the lines. A convex break gives a crest, a concave one gives a toe. The sign goes into the **kind** field, and 2.14 uses it to send a line into Top or Bottom.
- **A break without a drop is not a break.** A ridge of the evidence must gain the given relief drop in its neighbourhood, otherwise it is discarded before any thresholds. Centimetre noise never gains it, the neighbourhood of a crest always does. This is the main filter, and it is physical rather than statistical.
- **Forms are assembled by descent, not by proximity.** From the probe vertices of a crest a descent follows the flow directions to a toe, and the majority of probes picks the toe. Water from a crest runs exactly to its own toe, and on a curved wall with narrow berms this is the only way not to take the neighbouring bench.
- **A form is one toe with a set of crests at it.** The tracing cuts a long crest into pieces, all the pieces descend to the same toe, and the toe is written into the output once rather than once per piece.
- **The drop is measured within the probe base.** On a ten-metre bench with a three-cell base the drop reads three metres. The base is 8 cells by default, set it by the width of the face.
- **The significance threshold stays with the human.** A formal definition of a crest does not exist. What exists is the drop you are prepared to call a bench, and it differs on a quarry, on a road embankment and on a river bank.

Common mistakes

- **A DSM instead of ground.** Crowns and truck sides are breaks too, and the detector will find them. Classify before Isoliner.
- **A cell coarser than the bench.** A bench narrower than two cells does not exist in a raster. Make the cell smaller, parameters will not help here.
- **An empty output.** Look at the drop percentiles in the log. If the maximum is below the cut-off, the cut-off is to blame.
- **A line has glued two benches.** Visible in the 2.14 log as a low share of agreeing probes, and such a line goes into the unpaired. A higher drop cut-off or cutting the line manually helps.
- **A gentle junction at the toe.** Where a face passes smoothly into the base, the evidence is weak and the toe may not be found at all. There is physically no break there, so this is a limit of the method rather than a failure.
- **The descent did not reach a toe.** The path limit is too small, or the crest is false.